Basmala Nassar

B.SC. COMPUTER ENGINEERING · TECHNION · SOFTWARE & HARDWARE

Haifa, Israel · basmalan@campus.technion.ac.il · basmalh2004.github.io

Computer Engineering student at the Technion working across the whole stack: **systems software in C, C++ and Python**, **x86-64 binary optimization**, **computer-architecture simulation** and **digital design in SystemVerilog**. I drive every project to a fully passing test suite and debug from application code down to the binary and the hardware, with daily Linux fluency.

> **Expected graduation:** October 2028 · **Looking for:** a student position in software or hardware engineering, 2–4 days/week alongside studies · **Location:** Haifa / Tel Aviv, Israel

EDUCATION

B.Sc. Computer and Software Engineering, Technion

Nov 2023 – Oct 2028 (expected)

Andrew & Erna Viterbi Faculty of Electrical and Computer Engineering, Haifa · GPA 81.6

- **Selected grades:** Basic Software Lab (98) · Software Systems, C/C++ (82) · Digital Systems & Computer Structure (82) · Introduction to CS, C (81) · Operating Systems (77)
- **Software coursework:** Data Structures & Algorithms · Operating Systems · Computer Networks (TCP/IP) · Dynamic Binary Translation & Optimization · Correctness Guarantees for Software (SMT, abstract interpretation)
- **Hardware coursework:** Computer Architecture · Electronic Circuits (analog & digital VLSI) · Semiconductor Devices (PN, MOS, BJT) · High-Speed Digital Lab (transmission lines, reflections, jitter)

PROJECTS

Profile-Guided Binary Optimizer for x86-64

C++ · x86-64 · Intel Pin · Linux

Dynamic Binary Translation & Optimization · team of two · goal: beat the native run (no Pin) by >5%

- Pin probe-mode tool that profiles a running program, then builds an optimized copy of its hot code and migrates execution to it; **bzip2 runs 15.1% faster than native** on the course VM, faster in 7 of 7 runs
- Optimizations: huge pages for the new code, relocated switch jump tables, hot-loop alignment, cheaper profiling counters, and **stack-slot load forwarding** that serves `-O0` stack reloads from free registers
- Root-caused a crash in static glibc startup: the new code region collided with the randomized heap start; fixed by placing it above the real program break. Every output checked byte-for-byte against native

Systolic-Array Convolution Accelerator (VLSI Project)

RTL design · FSMs · Python

Technion VLSI Lab · team of two · 2026/27, in progress

- Configurable accelerator IP for CNN convolution layers: caches, AXI memory port, systolic MAC array, merge/quantize and pooling units, with a controller that overlaps loading and computing
- Designed the state machine of every unit, then wrote a **cycle-by-cycle Python model** of them to measure array utilization before RTL; its findings raised 1×1-layer utilization from **40.7% to 95.8%**

Computer Architecture Simulators

C · C++ · Python · Linux

Computer Architecture (046267) · each graded against a hidden automated test suite

- **Branch predictor** — BTB with local/global history and GShare/LShare XOR indexing into per-branch or shared FSM tables; **458/458 tests**
- **Two-level cache** — L1/L2 with LRU, write-back / write-allocate policies and average-access-time statistics; **803/803 tests**
- **Dataflow analyzer** — instruction-dependency DAG and critical-path depths for out-of-order execution; **500/500 tests, Valgrind-clean**
- **Branch profiler** — Pin tool ranking hot branches in real ELF binaries; traced a SIGSEGV under GDB to a rewritten lazily-bound PLT stub and fixed it

Network Interface Card Simulator

C++ · OOP · Make

Introduction to Software Systems (044101)

- L2/L3/L4 packet classes behind one abstract interface, built by a **factory method**; each layer validates its checksum, MAC or TTL and hands the inner packet down
- Routes packets by IP mask to the receive queue, transmit queue or open-port memory, rewriting source IP, TTL and checksum; **leak-free under Valgrind**

Additional Projects

SystemVerilog · C · Python

- **Pac-Man on FPGA**, EE Lab 1A (grade 90) — VGA controller, sprite rendering and FSM game logic in SystemVerilog; verified in ModelSim, then synthesized and debugged on the board
- **Linux shared library** — `libgrades.so` from a hand-written Makefile (`-fPIC`, `-shared`), ADT behind an opaque iterator, leak-free · **Numerical PDE solver** in Python/NumPy

SKILLS

Languages	**C**, **C++**, **Python** (NumPy, Matplotlib), **SystemVerilog**, x86-64 assembly, MATLAB
Software	Data structures & algorithms, OOP, operating systems (processes, memory, concurrency), TCP/IP networking, shared libraries & dynamic linking, ELF
Hardware	RTL & FSM design, computer architecture (caches, branch prediction, pipelines), FPGA bring-up, signal integrity, lab instruments (oscilloscope, signal generators)
Tools	Git, Linux, GCC/Make, GDB, Valgrind, Intel Pin, ModelSim, Quartus, Vivado
Spoken	Arabic (native) · Hebrew (fluent) · English (fluent)

TEACHING & LEADERSHIP

- **Private physics tutor** (self-employed, 2023–present) — mechanics, E&M and waves for high-school and first-year university students
- **Mentor, Technion Equal Opportunities Program** — guided first-year students through course registration and study strategy